

TASHTYP, Russian Federation

WMO: 299560

Lat: 52.800N

Lon: 89.904E

Elev: 1476

StdP: 13.93

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.7	-26.0	-36.5	0.7	-31.6	-30.8	1.0	-25.8	19.2	14.9	16.8	16.2	0.5	270	0.616

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.9	83.9	65.8	80.7	64.3	77.7	63.1	67.9	79.5	66.3	77.6	64.6	74.8	5.9	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.8	93.7	73.0	62.1	88.2	71.1	60.6	83.3	69.6	33.3	79.8	31.9	77.7	30.5	75.3	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.8	15.2	13.2	DB	-36.4	90.1	7.1	3.4	-41.5	92.5	-45.6	94.5	-49.6	96.5	-54.7	99.0	(4)
(5)			WB	-36.1	71.5	6.2	1.8	-40.5	72.8	-44.1	73.8	-47.6	74.8	-52.1	76.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	34.7	-0.7	5.5	22.6	39.2	50.3	61.5	64.7	60.5	48.5	35.6	19.8	6.5		
	DBStd	24.58	14.23	13.77	12.46	8.82	8.13	6.26	5.03	5.57	7.47	8.50	12.43	13.68		
	HDD50	6921	1570	1246	848	334	99	3	0	2	114	452	905	1348		
	HDD65	11182	2035	1666	1313	775	457	135	67	160	494	912	1355	1813		
	CDD50	1325	0	0	0	9	109	349	457	327	70	4	0	0		
	CDD65	115	0	0	0	0	2	32	59	21	1	0	0	0		
	CDH74	1554	0	0	0	4	122	479	587	320	42	0	0	0		
	CDH80	334	0	0	0	0	22	115	120	70	7	0	0	0		
	WSAvg	5.3	4.1	4.2	5.7	6.8	6.6	5.2	4.6	4.4	5.0	5.6	6.0	5.3		
	PrecAvg	20.8	0.6	0.4	0.6	1.6	2.6	3.4	3.3	3.0	2.0	1.4	1.1	0.7		
(15) Precipitation	PrecMax	31.3	1.2	0.9	1.8	4.3	6.8	8.8	6.1	5.1	4.1	3.5	3.1	1.4		
	PrecMin	14.5	0.2	0.0	0.1	0.3	0.9	1.2	1.7	1.5	0.7	0.4	0.3	0.2		
	PrecStd	3.4	0.3	0.2	0.4	1.1	1.3	1.7	1.1	1.1	0.8	0.8	0.6	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.0	37.3	56.4	72.4	83.6	88.2	87.5	86.2	79.9	64.8	47.9	34.8		
		MCWB	32.7	32.9	44.5	52.8	60.6	65.3	68.1	67.0	62.3	51.1	40.2	31.9		
	2%	DB	28.8	32.6	48.8	66.2	78.0	83.4	83.2	81.2	72.1	58.3	42.1	30.4		
		MCWB	26.2	28.9	40.5	49.7	57.9	64.7	66.8	64.7	58.0	47.8	37.2	27.7		
	5%	DB	24.2	28.9	42.9	60.9	73.2	79.7	80.2	77.2	67.0	53.2	38.2	28.1		
		MCWB	22.2	25.5	36.4	47.4	55.6	63.4	65.7	63.5	55.3	45.0	34.6	25.8		
	10%	DB	20.0	24.9	38.6	55.6	68.2	75.9	77.2	73.7	62.4	48.8	35.1	25.3		
		MCWB	18.0	22.1	33.6	44.9	53.4	62.0	64.6	61.7	53.3	42.3	32.2	23.0		
	0.4%	WB	33.0	33.5	44.8	53.9	62.7	68.5	71.2	69.0	63.3	52.1	41.1	32.6		
		MCDB	34.8	36.8	55.1	68.9	79.4	81.2	83.6	82.8	77.0	62.3	46.0	34.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	26.4	29.0	40.5	50.8	59.4	66.5	68.8	66.6	59.1	48.6	37.5	28.0		
		MCDB	28.6	32.4	47.9	63.9	74.8	79.6	80.0	78.5	69.5	56.6	41.1	30.1		
	5%	WB	22.4	25.6	36.7	48.2	56.8	64.8	67.0	64.5	56.6	45.5	34.9	25.8		
		MCDB	24.3	28.7	42.2	59.9	70.0	77.0	77.3	74.9	65.3	52.7	38.0	28.1		
	10%	WB	18.2	22.2	33.9	45.3	54.5	62.9	65.4	62.6	54.1	42.7	32.3	23.1		
		MCDB	20.0	24.9	38.4	55.0	66.2	73.8	74.8	71.9	61.1	48.4	35.0	25.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	17.2	19.5	20.6	21.3	22.9	22.7	19.9	21.0	20.7	17.3	13.5	15.3		
		MCDBR	17.3	20.6	24.6	33.3	35.5	30.8	26.5	29.3	30.5	27.1	16.0	14.5		
		MCWBR	15.6	17.6	16.6	19.0	17.6	14.1	11.9	14.3	16.5	17.1	11.9	13.0		
		MCDBR	17.3	20.4	23.2	31.0	29.7	26.1	22.8	26.8	27.2	25.7	15.1	14.5		
	5% WB	MCWBR	15.7	17.7	16.1	18.3	15.5	13.1	11.2	13.9	15.3	16.6	11.7	13.2		
(40) Clear-Sky Solar Irradiance	taub		0.280	0.316	0.364	0.408	0.407	0.396	0.404	0.391	0.355	0.346	0.320	0.287		
	taud		2.103	2.053	2.059	2.137	2.208	2.336	2.339	2.345	2.386	2.280	2.150	2.067		
	Ebn at Noon		215	242	255	260	267	272	267	264	260	233	196	184		
	Edh at Noon		23	33	41	43	42	38	37	35	30	27	22	20		
(44) All-Sky Solar Radiation	RadAvg		356	680	1096	1413	1642	1879	1731	1455	1057	596	316	233		
	RadStd		35	54	92	103	158	169	115	114	93	61	39	29		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page